Spectral Super-Resolution via Deep Low-Rank Tensor Representation

Renwei Dian, *Member, IEEE*, Yuanye Liu, and Shutao Li, *Fellow, IEEE*

***Abstract*— Spectral super-resolution has attracted the attention of more researchers for obtaining hyperspectral images (HSIs) in a simpler and cheaper way. Although many convolutional neural network (CNN)-based approaches have yielded impressive results, most of them ignore the low-rank prior of HSIs resulting in huge computational and storage costs. In addition, the ability of CNN-based methods to capture the correlation of global information is limited by the receptive field. To surmount the problem, we design a novel low-rank tensor reconstruction network (LTRN) for spectral super-resolution. Specifically, we treat the features of HSIs as 3-D tensors with low-rank properties due to their spectral similarity and spatial sparsity. Then, we combine canonical-polyadic (CP) decomposition with neural networks to design an adaptive low-rank prior learning (ALPL) module that enables feature learning in a 1-D space. In this module, there are two core modules: the adaptive vector learning (AVL) module and the multidimensionwise multihead self-attention (MMSA) module. The AVL module is designed to compress an HSI into a 1-D space by using a vector to represent its information. The MMSA module is introduced to improve the ability to capture the long-range dependencies in the row, column, and spectral dimensions, respectively. Finally, our LTRN, mainly cascaded by several ALPL modules and feedforward networks (FFNs), achieves high-quality spectral super-resolution with fewer parameters. To test the effect of our method, we conduct experiments on two datasets: the CAVE dataset and the Harvard dataset. Experimental results show that our LTRN not only is as effective as state-of-the-art methods but also has fewer parameters. The code is available at https://github.com/renweidian/LTRN.**

***Index Terms*— Adaptive low-rank prior learning (ALPL), canonical-polyadic (CP) decomposition, multidimensionwise multihead self-attention (MMSA), spectral super-resolution.**

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Renwei Dian is with the School of Robotics, Hunan University, Changsha 410082, China (e-mail: drw@hnu.edu.cn).

Yuanye Liu is with the School of Robotics and the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Hunan University, Changsha 410082, China (e-mail: yuanye_liu@hnu.edu.cn).

Shutao Li is with the College of Electrical and Information Engineering and the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Hunan University, Changsha 410082, China (e-mail: shutao_li@hnu.edu.cn).

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I. Introduction

HYPERSPECTRAL imaging, a combination of imaging and spectroscopic techniques, is capable of recording the spectral responses of a target in narrow bands of tens to hundreds of bands. A hyperspectral image (HSI) can be seen as a 3-D data cube, in which the spectral responses of different bands are stacked. An HSI can be expressed as $\mathcal{X}(x, y, \lambda)$, in which x and y represent the 2-D spatial coordinates and λ represents the 1-D spectral coordinate. Thanks to the detailed spectral information of HSIs, hyperspectral imaging is widely used in the fields of classification [1], [2], [3], [4], [5], medical image processing [6], [7], [8], food safety [9], [10], environmental monitoring [11], and so on.

Nonetheless, high spectral resolution imaging comes at the expense of temporal and spatial resolution [12]. Specifically, since the spectrum is divided into more segments, less energy is contained in each segment. Therefore, the photosensitive area of each photosensitive spot needs to be increased to obtain sufficient energy. Meanwhile, the size of the imaging component is limited resulting in lower spatial resolution. What is more, the photosensitive arrays in imaging equipment are at most 2-D, so an imaging device acquires only the spectral response values of two dimensions simultaneously resulting in the need for more exposure time to obtain a 3-D data cube. Most current hyperspectral imaging systems in the market cannot balance spectral resolution and spatial resolution and are not suitable for dynamic scenes. Therefore, hyperspectral imaging may have some limitations in practical applications.

With the advantages of hyperspectral imaging, many researchers have begun to improve the hyperspectral imaging system. However, these methods rely on expensive imaging devices, so many researchers attempt to obtain high-resolution HSIs via software algorithms These HSI reconstruction methods can be divided into two main categories: spectral–spatial fusion [13], [14], [15], [16], [17] and spectral super-resolution [18], [19], [20], [21]. For the former, researchers attempt to recover the HSI with high spatial resolution from an HSI with low spatial resolution and a multispectral image (MSI) or an RGB image. Since the requirement of the two images of a scene is demanding in a real environment, this method has limitations to some extent. In this article, we focus on another approach that obtains spectral information through the inverse mapping from RGB image to HSI, eliminating the need for an additional hyperspectral camera. These spectral details can offer more data support for distinguishing between various types and materials of objects. For instance, we differentiate between jadeite and Khotan jade using super-resolution algorithms. The RGB response values may vary due to lighting conditions, making it challenging to distinguish between them accurately. By employing spectral super-resolution algorithms, we acquire more spectral information, enabling precise identification. Similarly, in other classification and segmentation tasks, spectral super-resolution can utilize RGB images to provide additional spectral information. This information reflects the characteristics of different materials and substances, enhancing the ability to discern details. Compared to spectral–spatial fusion methods, this approach is more flexible and cheaper. Nevertheless, an RGB image may correspond to different HSIs. The conventional methods mainly use statistical principles to summarize some prior information and design sparse dictionaries or cardinal weighted combinations to complete the mapping from RGB images to HSIs. Unfortunately, hyperspectral datasets are very scarce in real-world scenarios, which makes it difficult to obtain complete prior information. Meanwhile, these artificial priors are related to the spectral response function, which varies by the imaging device.

The task of spectral super-resolution is an underdetermined problem, so it is difficult to obtain an accurate result. Fortunately, convolutional neural network (CNN) performs very well in various underdetermined problems, which also has been applied in HSI reconstruction. Plenty of CNN-based network frameworks have been designed ingeniously to improve the effects of spectral super-resolution, and these methods continue to refresh the quality evaluation metrics of spectral super-resolution. Although these CNN-based approaches achieve impressive results, most of them improve performance by increasing network complexity [22].

Taking into account of above problems, a novel low-rank tensor reconstruction network (LTRN) is presented for spectral super-resolution, which aims to exploit the low-rank prior of HSIs. Many research works [23], [24], [25], [26], [27], [28] have shown that taking full advantage of the low-rank characteristic of HSI improves the image restoration performance. We integrate the canonical-polyadic (CP) decomposition theory into deep learning approach to decompose low-rank tensors into sets of Kronecker basis vectors. Then, we only need to learn the Kroneker basis vectors through the neural network, which can produce the needed rank-1 tensors. Therefore, we extract the features from these one-dimension basis vectors instead of 3-D data. Specifically, we designed an adaptive low-rank prior learning (ALPL) module as the main structure of the encoder to generate the aggregated low-rank tensor. Our ALPL module uses an adaptive vector learning (AVL) strategy to get more accurate features. Besides, we propose the multidimensionwise multihead self-attention (MMSA) module, added into the ALPL, to promote the whole module to focus on the relevance of global context information in spatial and spectral dimensions. Furthermore, we design a simple feedforward network (FFN) as a subdecoder that is paired with the ALPL module. Our contributions are summarized as follows.

1) A novel LTRN is proposed for spectral super-resolution, which combines CP decomposition to make full use of the low-rank characteristics of HSIs.
2) We present a novel ALPL module, consisting of the MMSA module and AVL modules, which converts 3-D features into 1-D vectors and reduces the computational cost. The MMSA can capture long-range dependencies in the row, column, and spectral dimensions, respectively. The AVL module can improve the flexibility of feature extraction, so as to improve the feature representation ability.
3) Experimental results on two public datasets show that our method obtains high-quality HSIs with lower computational cost compared to the state-of-the-art methods. Furthermore, we validate the effectiveness of LTRN on real-world data.

II. Related Work

The research for spectral super-resolution can be traced back to the early work [29], [30], [31] that used principal component analysis (PCA) or k-singular value decomposition (K-SVD) to obtain sparse expressions of spectra for spectral recovery. For example, Parmar et al. [30] proved that nonnegativity constraint KSVD (KSVD-NN) is more excellent than KSVD and PCA in the sparse representation of spectra. They used gradient projection for sparse reconstruction (GPSR) [32] algorithm to learn a global spectral dictionary. Arad and Ben-Shahar [33] employed the K-SVD method to establish a sparse hyperspectral dictionary D_h under sparse constraints and used the orthogonal match pursuit (OMP) to project it into the RGB sensor space to obtain an RGB dictionary D_{rgb}. To decrease the number of atoms for the dictionary, Fu et al. [34] learned multiple dictionaries according to the clustering results in the training set. A different study introduced the radial basis function network to approximate the nonlinear mapping from RGB images to HSIs. The authors used the linear least squares method to solve the lighting parameters in combination with the imaging model and used the PCA basis function to make it fall into a defined subspace. Besides, they used extra white balance to make their model adaptable to other lighting conditions. However, these methods have not achieved outstanding results due to the ill-posedness of spectral super-resolution and their limited nonlinear modeling capabilities.

Recently, deep learning approaches have become the focus of spectral super-resolution due to their outstanding performance. In the early exploration, Galliani et al. [35] used a variant of Tiramisu, which is a semantic segmentation architecture based on Densenet [36]. Later, many conventional convolutional methods were proposed, such as in work [37], [38], and so on. However, these works mainly focused on designing deeper or wider network structures, ignoring the intrinsic properties of spectral data and other important influencing factors. Arad and Ben-Shahar [39] were the first to realize that the spectral function has a great impact on the accuracy of HSI reconstruction. Then, considering the influence of the spectral function, Fu et al. [40] presented a camera

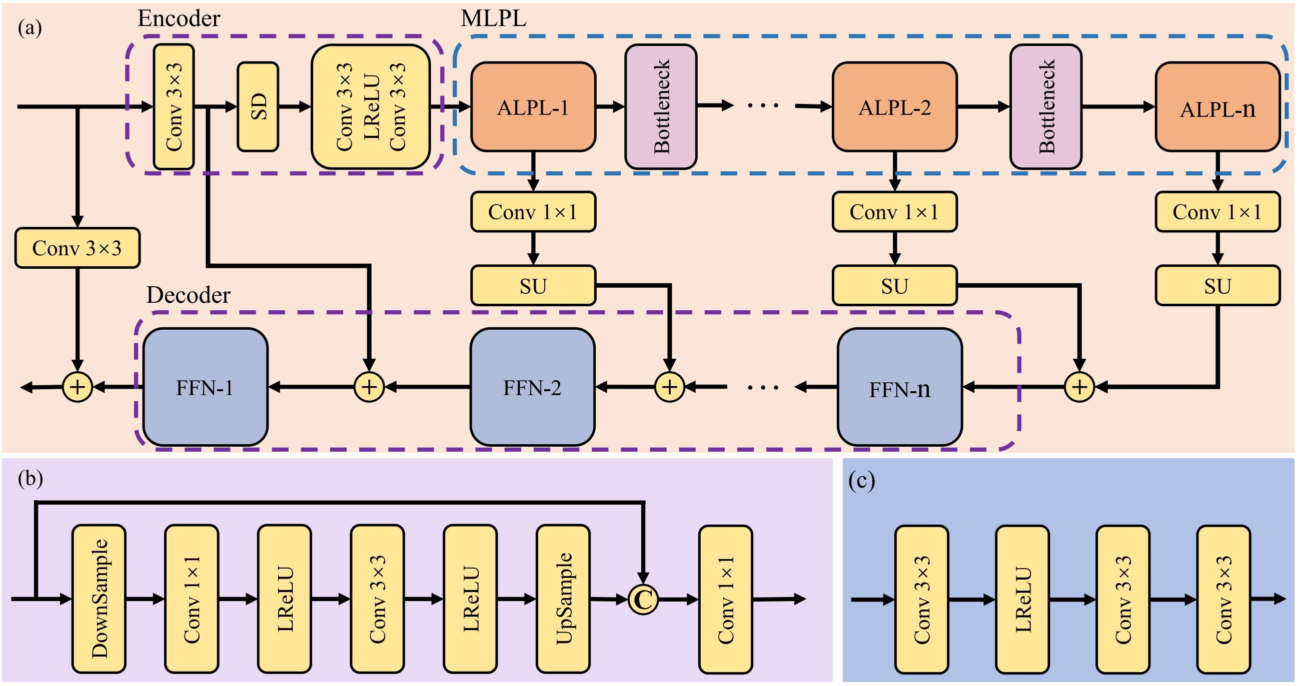


Fig. 1. Overall architecture of our LTRN. (a) LTRN consists of three parts: encoder, MLPL module, and decoder. (b) Bottleneck. (c) FFN. SD and SU represent spatial downsampling and spatial upsampling, respectively.

spectral selection layer and a recovery network composed of a spectral CNN for nonlinear spectral mapping and a spatial CNN for spatial similarity. Except for the previous methods, a loss function was proposed by Li et al. [22], which takes into account the camera spectral sensitivity. Fubara et al. [41] used a U-net network to obtain the weights of basis functions and used a matrix variable as the basis functions. In particular, they designed an unsupervised learning architecture to solve the problem of no ground-truth hyperspectral data in some cases, which required knowing the spectral response function. However, most of these methods treat the channelwise features equally, hindering the better discriminative ability for different types of features [42]. Then, attention mechanisms are gradually introduced into spectral super-resolution work. For example, Zhang et al. [42] designed a residual-in-residual (RIR) structure and introduced the channel attention mechanism to capture distinguishing features. Dai et al. [43] proposed a better attention mechanism based on second-order statistics of features, which can capture high-frequency information. Liu et al. [44] proposed a CNN-based method named SGARDN, which integrated residual block and spectral attention mechanism. They grouped the spectrum according to the covariance matrix and extracted features within and between groups in turn. Li et al. [45] designed a dual second-order attention mechanism and incorporated camera spectral sensitivity constraints into the loss function. In addition, some works have improved interpretability by designing convolutional networks based on optimization models. For instance, He et al. [46] integrated deep learning methods with optimization algorithms and guided spectral grouping using spectral response functions. Similarly, Dian et al. [47] regularized spectral super-resolution using spatial and spectral priors from HSI and established an imaging-model-guided spectral super-resolution optimization framework.

III. Our Proposed Method

A. Overall Architecture

Fig. 1(a) depicts the proposed LTRN that consists of three parts: feature encoding, low-rank prior learning, and feature decoding.

We denote an RGB image as $\mathcal{X} \in R^{w\times h\times 3}$, which is the input of the proposed LTRN. We first use one individual convolution for preliminary encoding

$$\mathcal{F}'_{\mathrm{PE}} = \Phi_{\mathrm{PE}}(\mathcal{X}) \tag{1}$$

where $\Phi_{\mathrm{PE}}(\cdot)$ represents a 3×3 convolutional layer with output channel number C. In this article, we set $C = 32$. To ensure that the subsequent modules can fully learn the correlation and low-rank prior between spectral channels, C must be more than the number of bands for HSIs. If $\mathcal{F}_{\mathrm{PE}}$ is directly fed into the ALPL module, it may result in a huge computational cost, which is contrary to our original design intention. Therefore, we exploit a 4×4 convolutional layer to achieve spatial downsampling with stride $= 2$. Besides, a deep encoder, consisting of two 3×3 convolutional layers and a LeakyReLU activation function, encodes the feature after dimensionality reduction. This process is formulated as

$$\mathcal{F}_{\mathrm{DE}} = \Phi_{\mathrm{DE}}(\Phi_{\mathrm{SD}}(\mathcal{F}_{\mathrm{PE}})) \tag{2}$$

where $\Phi_{\mathrm{DE}}(\cdot)$ and $\Phi_{\mathrm{SD}}(\cdot)$ denote the deep encoder and the downsampling function. Then, $\mathcal{F}_{\mathrm{DE}}$ is fed to the multistage low-rank prior learning (MLPL) module, and we can obtain encoded features at different stages

$$[\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_n] = \Phi_{\text{MLPL}}(\mathcal{F}_{\text{DE}}) \tag{3}$$

where $\Phi_{\text{MLPL}}(\cdot)$ denotes the MLPL module composed of n ALPL modules. Then, the decoder, composed of n FFNs, restores the HSI by making full use of $[\mathcal{F}_{\text{EN}1}, \mathcal{F}_{\text{EN}2}, \ldots, \mathcal{F}_{\text{EN}n}]$. Usually, deep networks lose the fine-grained features of the image. To compensate for the details lost during encoding and low-rank prior learning, we introduce a global residual connection

$$\mathcal{Y} = \Phi_{\text{DE}}(\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_n) + \text{Conv}(\mathcal{X}) \tag{4}$$

where $\Phi_{\text{DE}(\cdot)}$ represents the decoder function. $\text{Conv}(\cdot)$ is a 3×3 convolutional layer extracting fine-grained features directly from the input RGB image $\mathcal{X}$.

It is worth emphasizing that every ALPL module is followed by a bottleneck module, except for the last one. We use $\mathcal{F}_i$ to represent the output of the ith ALPL module, and then, we can obtain

$$\mathcal{F}_i' = \Phi_{\text{Bneck}}(\mathcal{F}_i) \tag{5}$$

where $\mathcal{F}_i'$ is the output of the ith bottleneck and the input of the next ALPL module, and the $\Phi_{\text{Bneck}}(\cdot)$ denotes the bottleneck structure. As shown in Fig. 1(b), two pointwise convolutional layers are involved in the bottleneck block, both of which aim to enhance the modeling for spectral properties. The downsampling operation is also a strided 4×4 convolutional layer, which can halve the spatial dimension size of the feature tensor and double the spectral dimension size. The upsampling operation is a deconvolutional layer to restore the size of each dimension of the feature tensor, which corresponds to the downsampling operation.

B. ALPL Module

Tensor decomposition mines implicit relationships in data according to which we can represent the original tensor in terms of low-dimensional tensors. The common decomposition methods are CP decomposition, Tucker decomposition, tensor train decomposition [48], and tensor ring decomposition [49]. In this article, we design the ALPL module based on the principle of CP decomposition, which can convey low-rank information about HSIs. Before introducing the ALPL module, we explain the concepts related to CP decomposition.

1) Kronecker Product: Given two second-order tensors $\mathcal{A} \in R^{m\times n}$ and $\mathcal{B} \in R^{p\times q}$, then the Kronecker product of $\mathcal{A}$ and $\mathcal{B}$ can be represented as

$$\mathcal{A} \otimes \mathcal{B} = \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{bmatrix} \tag{6}$$

where $\otimes$ denotes the Kronecker product operation.

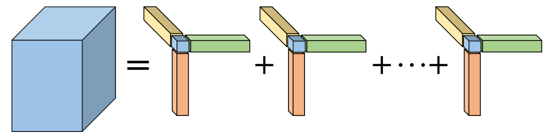


Fig. 2. CP decomposition visualization diagram of 3-D low-rank tensor.

2) CP Decomposition: Given an N-order tensor $\mathcal{X} \in R^{I_1\times I_2\times\cdots I_N}$, its CP decomposition can be written as

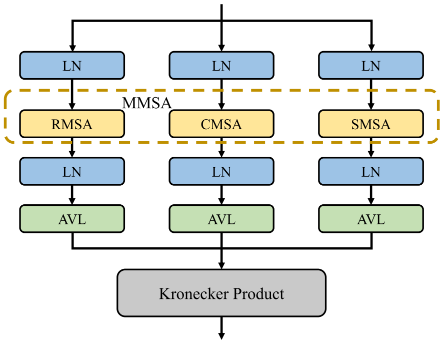
$$\mathcal{X} = \sum_{i=1}^{k} \omega_i v_{i1} \otimes v_{i2} \otimes \cdots \otimes v_{iN} \tag{7}$$

where $v_{in} \in R^{I_n}$ is a vector, called rank-1 Kronecker basis vector. $v_{i1} \otimes v_{i2} \otimes \cdots \otimes v_{iN}$ is the rank-1 tensor, and ω_i is the corresponding weight. k is the CP rank, a predefined number, which determines the number of rank-1 tensors.

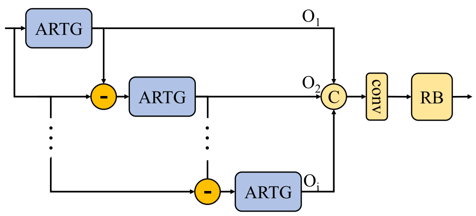
According to (7), CP decomposition can represent a low-rank tensor with k sets of vectors, which significantly reduces the dimensionality of the data. For spectral super-resolution, the data are 3-D tensors, whose CP decomposition is shown in Fig. 2. It is obvious that applying CP decomposition to the model can not only transfer low-rank prior information but also reduce the data dimension and has been applied to many image tasks. For example, Zhang et al. [50] have used CP decomposition for snapshot hyperspectral imaging and achieved impressive results. They propose tensor low-rank prior learning network (TLPLN) with global average pooling (GAP) to generate Kronecker basis vectors, which limits the ability of CP decomposition to convey information. To further exploit the ability of CP decomposition to transfer low-rank prior, we propose an ALPL module that can be seen as an upgrade of their TLPLN.

As shown in Fig. 3(a), we design an adaptive rank-1 tensor generation (ARTG) module consisting of three parallel branches. The key to CP decomposition is the accuracy of Kronecker basis vectors, so we use more flexible adaptive pooling to generate vectors instead of GAP, which is included in the AVL module. Due to the limitation of the field, CNN cannot capture global information. Therefore, in order to enable the model to capture long-distance dependencies, we introduce an MMSA mechanism. In addition, we introduce the layer normal operation before and after each MMSA module to accelerate the convergence of the entire model. In the last step, we can obtain the ran-1 tensor $\mathcal{Z}$ by performing the Kronecker product on the Kronecker basis vector.

From (7) and Fig. 2, it can be seen that the CP decomposition of a 3-D hyperspectral feature may result in k rank-1 tensors. To obtain these rank-1 tensors, we design the ALPL module based on the ARTG module and use the subtraction strategy to form the residual connection, which is consistent with literature [50], as shown in Fig. 3(b). Denote the input of ALPL module as $\mathcal{Z}_0$, and then, the first ARTG learns a rank-1 tensor $\mathcal{Z}_1$, which represents a frequency feature of the original feature, so $\mathcal{Z}_0 - \mathcal{Z}_1$ contains the features of other frequencies that the first ARTG module has not learned. The second ARTG module also learns a rank-1 tensor $\mathcal{Z}_2$ from $\mathcal{Z}_0 - \mathcal{Z}_1$, which is a new feature of another frequency, and so on. Then, the



(a)



(b)

Fig. 3. Architecture of (a) ARTG module and (b) ALPL module. The ALPL module contains the AVL module, the MMSA module, and layer normalization (LN).

ith ARTG module can be expressed as

$$Z_i = \Phi_{\text{ARTG}i}\left(Z_0 - \sum_{j=0}^{i-1} Z_j\right) \quad (8)$$

where Z_i is the output of the ith ARTG module. The number of ARTG modules in the ALPL module is determined by the CP rank of the original input tensor, which is predefined. Eventually, this module produces k rank-1 tensors that represent the different features of different frequencies.

The rank-1 tensors of the traditional CP decomposition have different weights $W \in R^{k\times 1} = [w_1, w_2, \ldots, w_k]$. However, this approach results in different spatial positions and spectral channels of the same rank with the same weight, which is inflexible and contrary to the fact that different spatial positions and spectral channels of HSIs have different importance. Moreover, this method ignores the correlation between features generated by features of different frequencies. Therefore, we use a 3×3 convolutional layer to fuse these rank-1 tensors

$$\mathcal{Y}_{\text{lr}} = \text{Conv}(\text{Concat}(\mathcal{Z}_1, \mathcal{Z}_2, \ldots, \mathcal{Z}_j)) \quad (9)$$

where $\mathcal{Y}_{\text{lr}}$ is the low-rank tensor and Concat$(\cdot)$ means stacking along rows or columns or dimensions. The low-rank tensor $\mathcal{Y}_{\text{lr}}$ is regarded as a 3-D map for the input characteristic tensor, which is generated by the attention vector in three dimensions. We then perform Hadamard product on input and $\mathcal{Y}_{\text{lr}}$. In order to refine the encoded feature, we design a simple residual block

$$\mathcal{F}_i = \Phi_{\text{RB}}(\mathcal{Z}_0 \times \mathcal{Y}_{\text{lr}}) \quad (10)$$

where $\Phi_{\text{RB}}(\cdot)$ denotes the residual block. As shown in Fig. 4, the residual block consists of four layers, in which the channel downsample operation and the channel upsample operation are both convolutional layers. The former shrinks the number of channels, and the latter enlarges the number of channels by the same factor consistent with r. The first two convolutional layers are followed by the ReLU activation functions, and the last two are followed by the LeakyReLU activation functions.

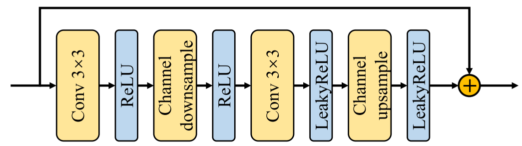


Fig. 4. Structure of the residual block. It is the last part of the ALPL module, which can further refine the feature details.

C. AVL Module

Similar to the channel attention mechanism, adaptive weights are more efficient than GAP operation, so we design an AVL module inspired by the AWCA [22] module to replace the GAP operation in the CP decomposition module, as shown in Fig. 5.

Specifically, we denote an intermediate feature tensor as $\mathcal{M} \in R^{C\times W\times H}$ whose number of channels is C and spatial size is $W \times H$. We utilize a convolutional layer and softmax layer to gain a 2-D map denoted as $W \in R^{1\times W\times H}$. Next, we reshape W and $\mathcal{M}$ to $R^{WH\times 1}$ and $R^{C\times WH}$, respectively. Then, we multiply $\mathcal{M}$ and W, which results in a vector v_{ap} with the length of C. The above processes are formulated as

$$v_{\text{ap}} = \Phi_{\text{AP}}(\mathcal{M}) \quad (11)$$

where Φ_{AP} represents the adaptive pooling operation. Each element of v contains the aggregation information of a channel, which can be regarded as the weight factor of the channel. In the last part of this module, we exploit two linear layers to do further learning for this vector. The final Kronecker basis vector is computed as

$$v = \Phi_{L2}(\Phi_{L1}(\mathcal{M})) \quad (12)$$

where $\Phi_{L1}(\cdot)$ and $\Phi_{L2}(\cdot)$ are the linear layers and corresponding activation functions. v is a Kronecker basis vector with length of C. Then, we can obtain three basis vectors with the length of C, W, and H by adjusting the input tensor to $C \times W \times H$, $W \times C \times H$, and $H \times W \times C$ as input of AVL, respectively. The AVL module takes full advantage of the expressive ability of CNN and extracts the features of HSIs more accurately in low-dimensional space.

D. MMSA Module

Recently, Cai et al. [51] have done meaningful work, which proposed spectralwise multihead self-attention (SMSA) with excellent performance. Inspired by this work, we proposed an MMSA module, which is similar to the SMSA mechanism. To simplify the network design work, we adopt the structure of SMSA to realize rowwise multihead self-attention (RMSA) and columnwise multihead self-attention (CMSA). As shown in Fig. 6, we take RMSA as an example to illustrate the

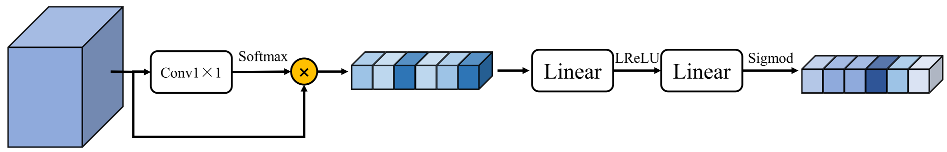


Fig. 5. AVL module. The AVL module compresses 3-D image features into 1-D vectors through adaptive pooling and adjusts them through linear layers. The final vector can be seen as the weight information of the corresponding dimension.

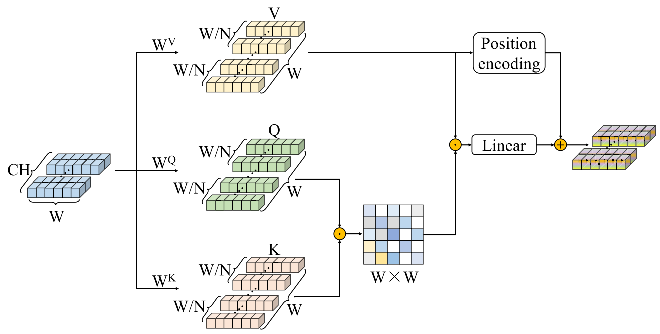


Fig. 6. MMSA module. The figure only uses RMSA as an example. This module can capture the global information of the row, column, and spectral dimensions of HSI features according to the different shapes of the input.

MMSA module. We suppose that the input tensor $\mathcal{F}_{\text{in}} \in R^{C\times H\times W}$ is the output of the previous module and reshape $\mathcal{F}_{\text{in}}$ to $R^{HC\times W}$ denoted as F. In common with the general self-attention mechanism, we first get value, query, and key from X_{in} through linear mapping denoted as V, Q, and $K \in R^{CH\times W}$, respectively,

$$
\begin{aligned}
V &= FW^V \\
Q &= FW^Q \\
K &= FW^K
\end{aligned}
\tag{13}
$$

where W^V, W^Q, and $W^K \in R^{W\times DN}$ are the network parameters that need to be learned in this structure. D is the dimension of a single head, and N is the number of heads. To save computational cost, we set that $DN = W$. We then calculate the similarity score by K and Q

$$C_j = \text{Softmax}\left(\lambda_j K_j^T Q_j\right) \tag{14}$$

where C_j is the self-attention score of the jth head, T represents transpose operation, and $\lambda_j \in R^1$ is an extra parameter to adapt to the variation of spectral density. Then, we can obtain the attention of the jth head by V as follows:

$$A_j = V_j C_j. \tag{15}$$

Finally, the outputs of N heads are spliced together and then transformed linearly through the fully connected layers. Before introducing position information, the outputs of N heads are spliced together and then mapped linearly through the fully connected layer

$$M = \text{Concat}(A_1, A_2, \ldots, A_j)W + \Phi_P(V) \tag{16}$$

where $M \in R^{C\times H\times W}$ is the output of the RMSA module and $\Phi_P(\cdot)$ is the positional encoding function. Φ_P consists of two convolutional layers, and there is a gaussian error linear unit (GeLU) activation between them. It should be emphasized that there are many reshaping operations to ensure that the dimensions of each intermediate data correspond to each other during the above process.

IV. Experiments

A. Datasets

To verify the effectiveness of the proposed network, we conducted experiments on two datasets. The first dataset is the CAVE[1] dataset [53] provided by Columbia University. These HSIs are taken by a refrigerated charge-coupled device (CCD) camera with a variable specification liquid crystal tunable filter. It has a total of 32 HSIs divided into five sections: stuff, skin and hair, paints, food and drinks, and real and fake. Each HSI has 31 spectral bands expressing a spectral response value between 400 and 700 nm at 10-nm intervals. We randomly select 20 HSIs as the training set, six HSIs as the validation set, and six HSIs as the test set.

The second dataset is the Harvard[2] dataset [54], provided by Harvard University. These HSIs are taken by a commercial hyperspectral camera (Nuance FX, CRi, Inc.) with an integrated liquid crystal tunable filter. It has a total of 50 HSIs of indoor and outdoor scenes under daylight illumination. Each HSI has 31 spectral bands expressing a spectral response value between 420 and 720 nm at 10-nm intervals. We randomly select 30 HSIs as the training set, ten HSIs as the validation set, and ten HSIs as the test set.

B. Compared Methods

We compare our model with advanced spectral super-resolution methods, including HSCNN [37], PriNet+ [52], HSRnet [46], AWAN [22], MST++ [51], SSRnet [47], and HSACS [45]. AWAN takes first place on the clean track of the NTIRE 2020 Spectral Reconstruction Challenge, and MST++ takes first place in the NTIRE 2022 Spectral Reconstruction Challenge. SSRnet and Prinet+ are two typical lightweight methods. We train all methods on the same dataset, ensuring that the training, validation, and test sets are consistent. These methods are all CNN-based methods, and we follow the best optimizer and learning rate given by the original author to train the corresponding model.

[1]http://www.cs.columbia.edu/CAVE/databases/multispectral/. The validated images are egyptian_statue_ms, fake_and_real_beers_ms, fake_and_real_food_ms, fake_and_real_lemons_ms, jelly_beans_ms, and thread_spools_ms. The tested images are fake_and_real_peppers_ms, fake_and_real_strawberries_ms, hairs_ms, pompoms_ms, real_and_fake_apples_ms, and stuffed_toys_ms.

[2]http://vision.seas.harvard.edu/hyperspec/. The validated images are imga1, imgb3, imgb4, imgb6, imgb7, imgd3, imge0, imgf3, imgf8, and imgh0. The tested images are img1, imga2, imga6, imgb0, imgc2, imgc8 imge2, imge5, imge7, and imgh2.

TABLE I

QUANTITATIVE METRICS FOR THE TEST METHODS ON THE CAVE DATASET AND THE HARVARD DATASET. THE RED BOLD FONTS MARK THE BEST EFFECT, AND THE BLUE FONTS MARK THE SECOND EFFECT

			CAVE				Harvard			
Method	Params (M)	FLOPS(G)	PSNR	RMSE	UIQI	SSIM	PSNR	RMSE	UIQI	SSIM
HSCNN	1.24	5.06	35.6299	5.3254	0.7683	0.9522	41.6920	3.6082	0.8536	0.9732
PriNet+	**0.57**	2.30	37.9629	5.2973	0.8550	0.9773	41.7098	4.0127	0.8503	0.9717
HSRnet	0.77	3.00	38.4459	4.6511	0.8527	0.9801	41.6952	3.5459	0.8571	0.9747
AWAN	21.36	88.44	**39.4262**	4.8245	0.8597	0.9798	42.2312	3.5034	0.8616	0.9769
MST++	1.62	**1.46**	38.5511	**4.6304**	**0.8731**	**0.9832**	**42.4756**	**3.2552**	**0.8623**	**0.9773**
SSRnet	**0.39**	1.58	38.6807	4.9824	0.8573	0.9794	42.1070	3.5370	0.8608	0.9760
HSACS	19.74	82.18	37.9112	5.4099	0.8389	0.9765	42.1360	3.3203	0.8586	0.9762
OURS	0.67	**0.85**	**39.7349**	**4.3095**	**0.8702**	**0.9832**	**42.4953**	**3.3112**	**0.8632**	**0.9770**

C. Quantitative Metrics

To quantitatively evaluate the network proposed in this article, we adopt four widely used image quality evaluation metrics, consisting of peak signal-to-noise ratio (PSNR), root mean square error (RMSE), universal image quality index (UIQI) [55], and structural similarity index (SSIM) [56]. PSNR is based on error-sensitive image quality evaluation by calculating the ratio of the variance of the signal to the noise. The UIQI is calculated from the variance and mean of the reconstructed HSI and reference HSI. SSIM measures the similarity of HSIs from three aspects: brightness, contrast, and structure. UIQI and SSIM evaluate the spatial similarity between the reconstructed image and the reference image from different perspectives. Among them, larger PSNR, UIQI, and SSIM indicate better quality of the reconstructed HSI. However, RMSE is the opposite.

D. Implementation Details

All methods are implemented on the PyTorch framework with a CPU of Intel i9-10920X and a GPU of NVIDIA RTX 6000. The number of ALPL modules n is set to 2, and the CP rank k is set to 3. The parameter optimizer uses Adam [57] with $\beta_1 = 0.9$ and $\beta_2 = 0.99$, and $\epsilon = 0.9$. The learning rate is initialized as 0.0005, and the cosine annealing scheme is adopted. We use the most basic L1 loss function as the optimization goal for the entire model. During the training process, we normalize all input images so that all image data are scaled between 0 and 1. Besides, we crop the image in a window of 64×64 with a stride of 32. For the CAVE dataset, we set the epoch to 1000 and the batch size to 32. For the Harvard dataset, we set the epoch to 200 and the batch size to 32 as well.

E. Performance Evaluation

We demonstrate the spectral super-resolution performance of different methods through quantitative evaluation using the assessment metrics introduced in Section IV-C. Table I records the results of different methods on the CAVE and Harvard datasets. On the CAVE dataset, our method achieves the best performance in terms of PSNR, RMSE, and SSIM metrics, with only UIQI slightly below the optimal result. Specifically, PSNR increases by 0.30 dB compared to AWAN, and RMSE decreases by 0.32 compared to MST++. On the Harvard dataset, our method and MST++ both achieve outstanding results, each with its own strengths. Table I also displays the parameters and floating point operations (FLOPS) of different methods. Although LTRN has more parameters compared to PriNet and SSRnet, its FLOPS are lower than those of PriNet and SSRnet. Moreover, our method has less than half the parameters and reduces FLOPS by 40% compared to the competitive MST++ when dealing with an input size of 64×64. Overall, our method reduces computational cost while ensuring the reconstruction quality of HSI.

F. Visual Evaluation

To show the effect of spectral super-resolution of different methods more intuitively, we select two test images from each dataset and present the corresponding reconstructed images for the 30th band and RMSE images for all bands, as shown in Figs. 7 and 8. For a clearer visual comparison, we enlarge specific details in the reconstructed images. By observing the enlarged details in the images, we conclude that MST++, SSRnet, and our method effectively restore the image details, making it difficult to distinguish the superiority among them. This illustrates the sophistication of these methods and ours, consistent with the conclusions drawn from quantitative evaluations. In the error maps, our method exhibits almost no highlighted error regions, and the overall colors are closer to the reference images. This leads to the conclusion that our method outperforms the others.

G. Experiments on Real Data

We experiment with real data that can further demonstrate the effectiveness of the proposed method. Fig. 9(a) shows the RGB image of jadeite and Khotan jade, which is taken by the Sony IMX296 sensor. Although the materials of the two jades are different, it is difficult for us to distinguish them from the RGB image. In order to distinguish them, we retrain LTRN on the CAVE dataset and perform spectral super-resolution on the RGB images. It is worth noting that we use the spectral response functions of the Sony IMX296 sensor to downsample HSIs, generating corresponding RGB images. The result is shown in Fig. 9(b), in which the spectral response of the Khotan jade is higher than that of the jadeite overall. This can help us distinguish them more accurately, which demonstrates the effectiveness of our proposed method.

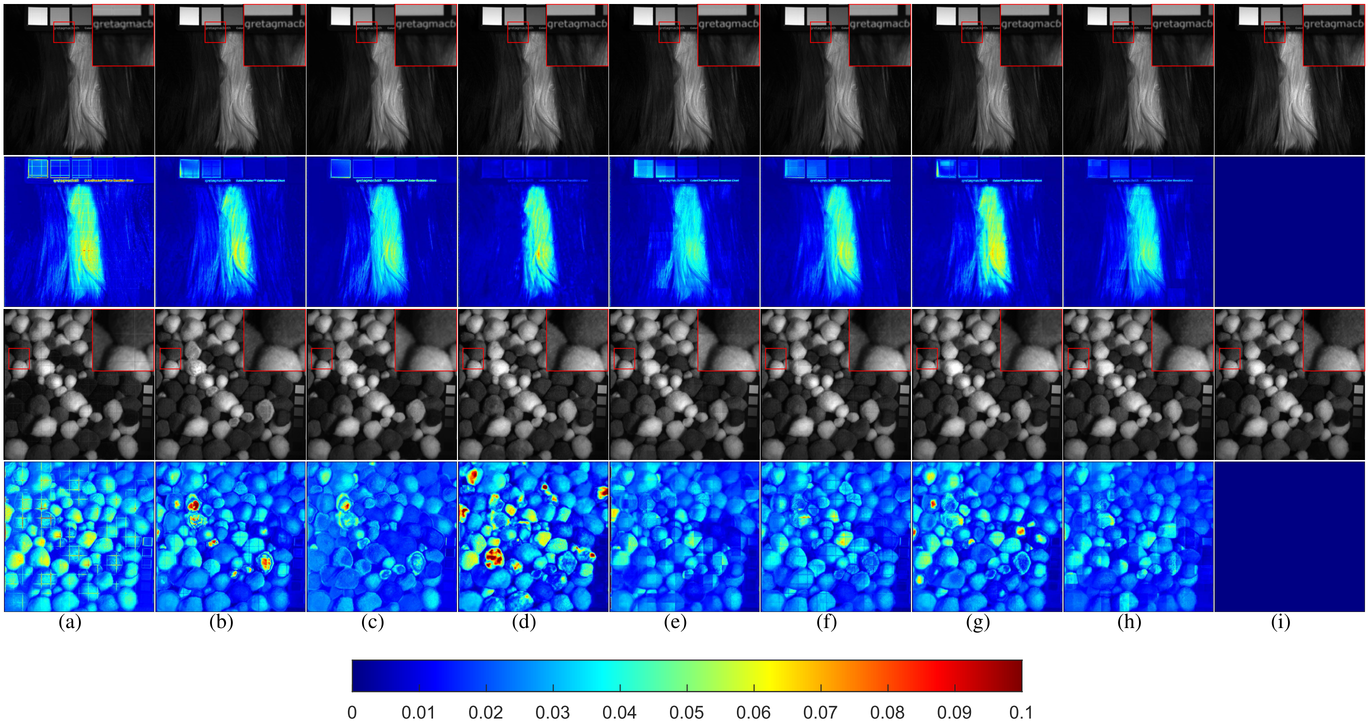


Fig. 7. Reconstructed images for the 30th band and RMSE images of hairs_ms and pompoms_ms (tow test images of the CAVE dataset). (a) HSCNN [37]. (b) PriNet+ [52]. (c) HSRnet [46]. (d) AWAN [22]. (e) MST++ [51]. (f) SSRnet [47]. (g) HSACS [45]. (h) LTRN (ours). (i) Reference.

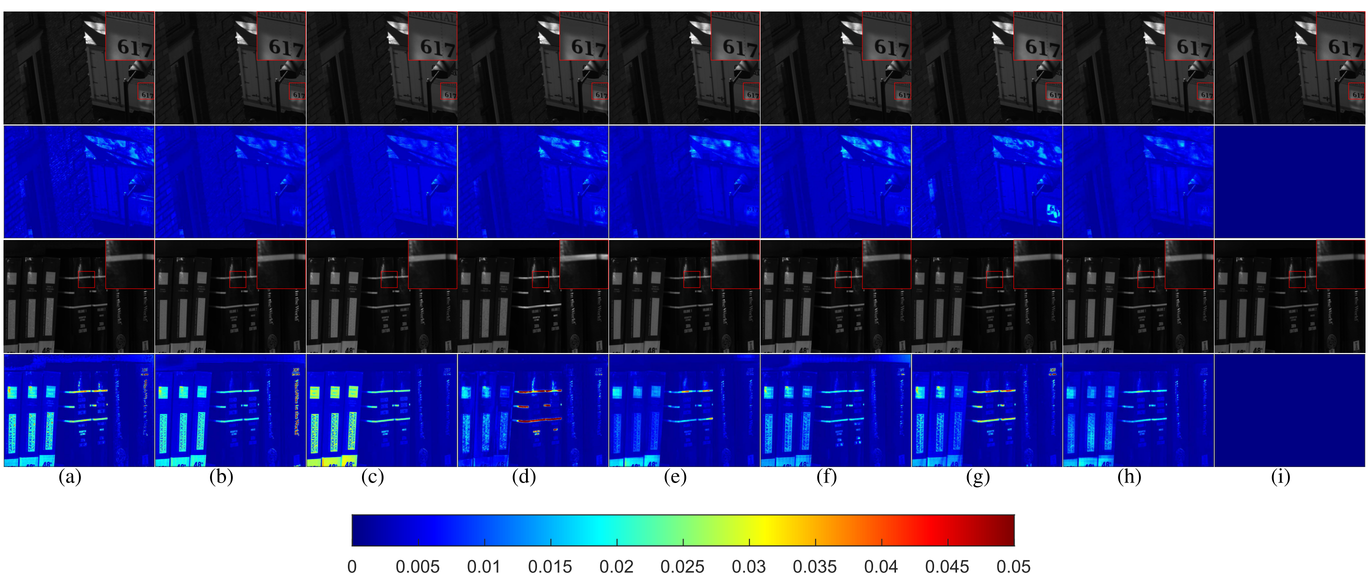
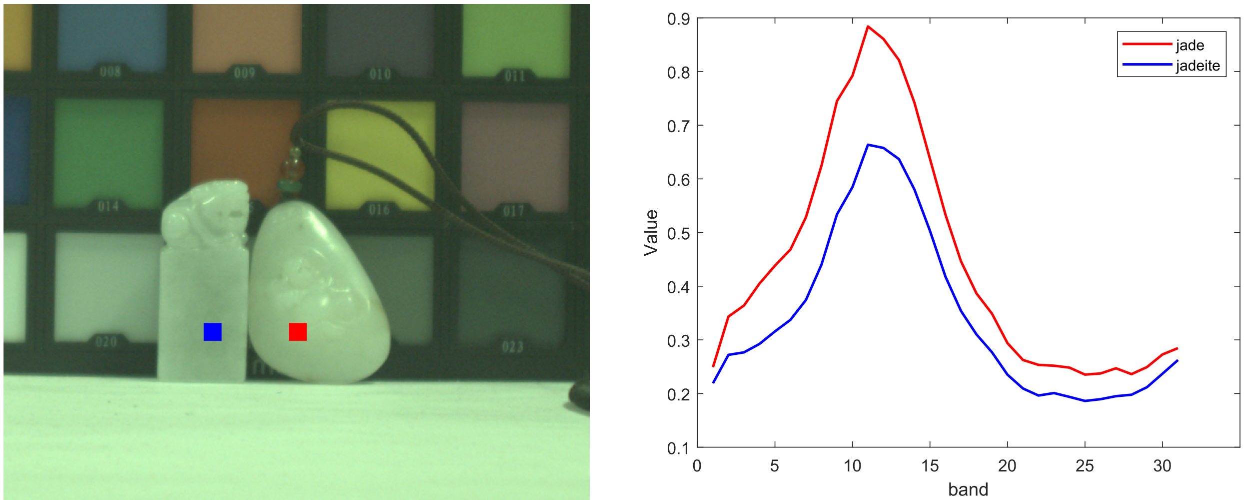


Fig. 8. Reconstructed images for the 30th band and RMSE images of imga2 and imgh2 (tow test images of the Harvard dataset). (a) HSCNN [37]. (b) PriNet+ [52]. (c) HSRnet [46]. (d) AWAN [22]. (e) MST++ [51]. (f) SSRnet [47]. (g) HSACS [45]. (h) LTRN (ours). (i) Reference.

H. Ablation Analysis

The most important module of the LTRN is ALPL, which consists of the MMSA module and AVL modules. In this section, we analyze their effects. Besides, we provide the performance of LTRN under different parameter settings.

1) Effects of Different Key Modules: To validate the effectiveness of each module, we first removed MMSA and used GPA instead of adaptive pooling, denoted as the "baseline." Then, on the basis of "baseline," we, respectively, added MMSA and adaptive pooling, denoted as "baseline + MMSA" and "baseline + AVL." Finally, MMSA and adaptive pooling are applied simultaneously, denoted as "baseline + MMSA + AVL." As shown in Table II, compared to the "baseline," the performance of "baseline + MMSA" does not improve and even causes worse evaluation metrics, such as UIQI, SSIM, and RMSE. However, "baseline + AVL" shows a slight advantage, slightly outperforming the baseline across all evaluation metrics. When both MMSA and adaptive pooling are introduced simultaneously, there is a noticeable



(a) (b)

Fig. 9. (a) RGB image that we took. (b) Spectral response curves of jadeite (blue) and Khotan jade (red) obtained by our method.

TABLE II
QUANTITATIVE METRICS OF LTRN WITH/WITHOUT AVL AND MMSA

	CAVE				Harvard			
Method	PSNR	RMSE	UIQI	SSIM	PSNR	RMSE	UIQI	SSIM
baseline	39.1508	4.6435	0.8652	0.9820	41.9978	3.7463	0.8596	0.9760
baseline+AVL	39.5704	4.2368	0.8662	0.9822	42.1609	3.4299	0.8618	0.9766
baseline+MMSA	39.1587	4.5860	0.8533	0.9808	42.1222	3.8075	0.8590	0.9759
baseline+AVL+MMSA	**39.7349**	**4.3095**	**0.8702**	**0.9832**	**42.4953**	**3.3112**	**0.8632**	**0.9770**

TABLE III
EFFECTS OF HYPERPARAMETER VALUES ON THE RECONSTRUCTION PERFORMANCE AT THE CAVE DATASET

Hyperparameters	Set	SAM	RMSE	UIQI	SSIM
	2	38.8686	4.5022	0.8484	0.9794
k	3	**39.7349**	**4.3095**	**0.8702**	**0.9832**
	4	38.9103	4.8137	0.8489	0.9791
	1	38.9986	4.4497	0.8503	0.9784
n	2	**39.7349**	**4.3095**	**0.8702**	**0.9832**
	3	39.1238	4.4406	0.8504	0.9792

TABLE IV
EFFECTS OF HYPERPARAMETER VALUES ON THE RECONSTRUCTION PERFORMANCE AT THE HARVARD DATASET

Hyperparameters	Set	SAM	RMSE	UIQI	SSIM
	2	42.1093	3.4496	0.8603	0.9764
k	3	**42.4953**	**3.3112**	**0.8632**	**0.9770**
	4	42.2577	3.5163	0.8595	0.9761
	1	42.1853	3.4396	0.8608	0.9764
n	2	**42.4953**	**3.3112**	**0.8632**	**0.9770**
	3	42.1856	3.4511	0.8595	0.9761

improvement in all evaluation metrics. Especially, the PSNR increased by 0.58 and 0.5 dB on the CAVE and Harvard datasets, respectively.

2) Hyperparameter Settings: The parameters of our model mainly refer to CP rank k and the number of ALPL modules n. We fix $k = 3$ and compare the results when $n = 1$, $n = 2$, and $n = 3$. The results, as shown in Table III, indicate that the model works best when $n = 2$. We fix $n = 2$ and compare the results of $k = 2$, $k = 3$, and $k = 4$, as shown in Table IV. It can be seen that when $k = 3$, the model works best. Therefore, the LTRN is configured with $k = 3$ and $n = 2$, which yields the best results.

V. CONCLUSION

In this work, we propose a spectral super-resolution method based on tensor decomposition, which relates HSIs with 3-D tensors and learns the low-rank prior of HSIs with the help of the powerful learning ability of deep learning methods. For low-rank tensor decomposition, we first use adaptive pooling to generate the decomposed vectors and learn the features of HSIs in the form of vectors through a fully connected network. Through tensor decomposition, problems in 3-D space are transformed into problems in 1-D space, which greatly reduces the calculation and storage requirements of the model. In the process of combining tensor decomposition with deep learning, we introduce MMSA to improve the accuracy of the tensor decomposition process. Multiple cascaded decomposition networks form the MLPL module, which is combined with a decoder formed by simple residual convolutional networks to complete the spectral super-resolution task. By comparing with multiple advanced methods, we prove the superiority of the proposed method. In addition, we verify the effectiveness of MMSA and AVL via ablation experiments.

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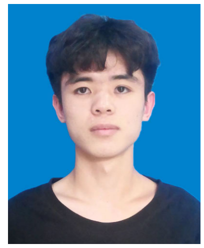
Renwei Dian (Member, IEEE) received the B.S. degree from the Wuhan University of Science and Technology, Wuhan, China, in 2015, and the Ph.D. degree from Hunan University, Changsha, China, in 2020.

From November 2017 to November 2018, he was a Visiting Ph.D. Student with the University of Lisbon, Lisbon, Portugal, supported by the China Scholarship Council. From August 2020 to July 2022, he held a post-doctoral position at the College of Electrical and Information Engineering, Hunan University, where he was awarded a distinguished post-doctoral fellowship. He is currently an Associate Professor with the School of Robotics, Hunan University. His research interests include hyperspectral image superresolution, image fusion, tensor decomposition, and deep learning.

Dr. Dian was a finalist for the Best Student Paper Award at the International Geoscience and Remote Sensing Symposium (IGARSS) 2018. He was awarded the Wu Wenjun Award for Outstanding Youth in Artificial Intelligence, the Fellowship of China National Postdoctoral Program for Innovative Talents, and an Excellent Doctoral Dissertation by the China Society of Image and Graphics. More information can be found on his homepage: https://sites.google.com/view/renweidian/.



Yuanye Liu received the B.S. degree from the School of Internet of Things Engineering, Jiangnan University, Wuxi, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Robotics, Hunan University, Changsha, China.

His research interests include hyperspectral image reconstruction and deep learning.

Shutao Li (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees from Hunan University, Changsha, China, in 1995, 1997, and 2001, respectively.

In 2001, he joined the College of Electrical and Information Engineering, Hunan University. From May 2001 to October 2001, he was a Research Associate with the Department of Computer Science, The Hong Kong University of Science and Technology, Hong Kong. From November 2002 to November 2003, he was a Post-Doctoral Fellow with the Royal Holloway College, University of London, London, U.K. From April 2005 to June 2005, he was a Visiting Professor with the Department of Computer Science, The Hong Kong University of Science and Technology. He is currently a Full Professor with the College of Electrical and Information Engineering, Hunan University. He has authored or coauthored over 200 refereed papers. His current research interests include image processing, pattern recognition, and artificial intelligence.

Dr. Li is a member of the Editorial Board of *Information Fusion* and *Sensing and Imaging*. He received the 2nd-Grade State Natural Science Award of China in 2019 and two 2nd-Grade State Scientific and Technological Progress Awards of China in 2004 and 2006. He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and the IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT.